

# Julius Zhang

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I am interested in identifying structures and developing new primitives for practical problems.

## EXPERIENCE

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| <b>Columbia University</b> <ul style="list-style-type: none"><li>Conduct research and attend seminars in the Department of Mathematics.</li><li>Homotopy theory, arithmetic geometry, cryptography, security, and other topics.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Researcher, Jun. 2026 - Present    |
| <b>a16z</b> <ul style="list-style-type: none"><li>Building <a href="#">Jolt</a>, open-source zero-knowledge virtual machine in Rust.</li><li>Implemented and benchmarked <a href="#">torus compression</a>; proved optimality with a conceptual reframing via Hilbert 90.</li><li>Audited proof systems security: verify correctness of <a href="#">elliptic-curve optimizations</a> and formalize folklore results into rigorous <a href="#">proofs</a>.</li><li>Evaluated lattice-based cryptography parameters against known attacks under various security regimes and performance constraints.</li><li>Performance engineering across the proving stack, including sum-check optimization.</li></ul> | Researcher, Apr. 2025 - Present    |
| <b>Millennium</b> <ul style="list-style-type: none"><li>Convex optimization and robust statistical estimation for real-time trading systems.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Researcher, Jun. 2024 - Apr. 2025  |
| <b>Stellar Development Foundation</b> <ul style="list-style-type: none"><li>Byzantine fault tolerance, C++, <a href="#">stellar-core</a>.</li><li>Improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.</li></ul>                                                                                                                                                                                                                                                                                                                                                           | Consultant, Jun. 2023 - Sept. 2023 |
| <b>Distributed Systems and TCS Research, Stanford University</b> <ul style="list-style-type: none"><li>Under David Mazières, research in general open membership Byzantine fault tolerance protocols (<a href="#">Github</a>), with applications to certificate authority management and automatic password recovery with hardware security modules.</li><li>Research PCP under Aviad Rubinfeld, reconstructed proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.</li></ul>                                                                                                                                                                                  | Researcher, Sept. 2023 - Jun. 2024 |
| <b>Topology Research, Stanford University</b> <ul style="list-style-type: none"><li>Floer homotopy theory under Mohammed Abouzaid; related the complex cobordism Gysin map to the Grothendieck residue symbol and derived a recursive formula for Gromov–Witten quantum product corrections.</li><li>Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.</li></ul>                                                                                                                                                                                                                                | Researcher, Jul. 2020 - May 2022   |

## SELECTED PAPERS AND MANUSCRIPTS

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| <b>Julius Zhang.</b> <i>The Relative Trace-Zero Subgroup of the Barreto-Naehrig Curves</i> , Cryptology ePrint Archive, 2026/1311. <a href="#">[PDF]</a>         |  |
| <b>Julius Zhang.</b> <i>Complex Cobordism and Formal Group Law</i> , Manuscript (Undergraduate Honours Thesis), Stanford University, 2024. <a href="#">[PDF]</a> |  |

## TALKS

- $p$ -Ordinary Cohomology for  $SL(2)$  over Number Fields**, Hida Theory Seminar, Columbia University, 2026.
- Construction of Seiberg-Witten-Floer stable homotopy types**, Homotopy Theory and Manifolds Seminar, Columbia University, 2026.

## EDUCATION

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|--------------------------------------------------------------------------------------------------------|------------------------|
| <b>Courant Institute of Mathematical Sciences, New York University</b> , computer science <i>PhD</i> . | starting Sept. 2026    |
| <ul style="list-style-type: none"><li>Verifiable systems and theoretical computer science.</li></ul>   |                        |
| <b>Stanford University</b> , computer science <i>M.S.</i> and mathematics <i>B.A.</i>                  | Sept. 2019 - Jun. 2024 |
| <ul style="list-style-type: none"><li>General Wang Yaowu Scholarship.</li></ul>                        |                        |

## SKILLS

**Computer Science:** Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

**Mathematics:** number theory, algebraic geometry, symplectic topology, homotopy theory, Floer theory, gauge theory, concentration inequalities.